28. Explicit finite-time action confinement from a normal form

Pure Thought research portfolio

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28. Explicit finite-time action confinement from a normal form

Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A system need not move on a perfectly orderly orbit to remain near its initial state for a very long time. Small disturbances can produce only a slow drift in certain quantities. This problem asks for an explicit guarantee: how far can those quantities change, and for how long is that bound valid? It focuses on a finite time horizon rather than proving that a special motion persists forever.

Key terms

- **Near-integrable Hamiltonian** — An energy function consisting of a well-understood integrable part plus a small perturbation.
- **Action** — A coordinate that is constant in the integrable reference system and may drift when a perturbation is added.
- **Confinement bound** — A guaranteed maximum change of the specified actions over a stated time interval.
- **Normal form** — A transformed expression of the equations separating manageable terms from a remainder.
- **Canonical transformation** — A change of variables preserving the Hamiltonian structure of the dynamics.
- **Remainder** — Terms not absorbed into the simplified normal form; bounding their size helps bound drift.
- **Resonance** — An integer relation among frequencies that can change how perturbations affect the motion.
- **Nekhoroshev theorem** — A theorem giving very long action-confinement times for small perturbations under specified regularity and nondegeneracy assumptions.
- **Quasi-convexity** — A particular curvature condition on the integrable energy function used by some such theorems.
- **Dimensional time** — A time expressed in declared physical or model units, rather than an unexplained numerical factor.

Why this matters

Finite-time guarantees answer a question that simulations alone cannot settle beyond their run length. They also distinguish a valid stability argument from an impressive-looking exponential formula with missing constants or units. Understanding slow drift is relevant to the mathematical study of planetary and other nearly integrable systems.

What progress would mean

A useful first result could be a modest but fully justified confinement time. Improving that time with controlled remainders would be real progress; asserting billions of years would require an appropriate physical model and every conversion and bound needed to support it.

Core question

For a specified analytic Hamiltonian and action domain, what rigorously computable action excursion and confinement time follow from a certified normal form or quantitative Nekhoroshev theorem?

Scope and assumptions

Use H=h(I)+ϵf(I,θ), h=ω·I+|I|^2/2, in two or three degrees of freedom on a compact action box with an explicit complex extension. Supply ω, a finite Fourier polynomial f, and rational ϵ. Bound the gradient away from zero where the selected theorem requires it.

The initial target is a valid dimensional time bound, even if modest. Planetary and post-Newtonian models are extensions requiring their own reductions and hypotheses.

Mathematical target

For a quasi-convex theorem verify its actual condition, for example v^{T}D^{2}h(I)v≥m|v|^2 for v⊥∇h(I), plus the theorem's gradient, domain and analyticity conditions. A positive Hessian determinant alone is insufficient.

Alternatively construct a canonical Φ with H∘Φ=Z(J)+R(J,θ), bound |I−J|≤η and ||∂_θR||≤r on a validated domain. Then |I(t)−I(0)|≤2η+r|t| while the transformed orbit remains in that domain. If resonant Z depends on angles, include its action drift or restrict the claim to conserved/protected components.

Required outputs and proof obligations

Export normal-form order, transformations, inverse/domain bounds, Fourier tails, r, η, and a domain-containment proof for the claimed interval. A Nekhoroshev result must include all constants and its smallness threshold in T=C exp(c $\epsilon\hat{}$−a) with stated units; do not choose a from a desired time target.

Validation and rejection controls

Verify integrable ϵ=0 behavior and a low-order symbolic normal form. Check symplecticity with validated remainder control. Confirm the action bound stays inside the certified domain. Independently recompute exp(10^0.3)≈7.35: it is a dimensionless factor, not 10^{13} years.

Milestones and research extension

Baseline: one explicit finite-time confinement certificate. Strong: optimized truncation with a stronger bound or full quantitative quasi-convex theorem application. Research extension: exponential scaling across a parameter family; do not infer actual solar-system stability from ϵ alone.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Zhang and Zhang, Improved stability for analytic quasi-convex nearly integrable systems — Theorem-specific quasi-convex exponential stability results.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$H \circ \Phi = Z(J) + R(J,\theta), \qquad |I - J| \le \eta, \qquad \|\partial_\theta R\| \le r,$$

$$|I(t) - I(0)| \le 2\eta + r|t|.$$

This estimate assumes the angle-independent normal form shown and a proved domain-containment interval. Include resonant drift if Z depends on angles. A quantitative Nekhoroshev theorem instead requires all its constants, hypotheses and threshold in $T = C\exp(c\epsilon^{-a})$, with units supplied by C. The number $\exp(10^{0.3}) \approx 7.35413$ has no unit of time.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — incorrect steepness (30–31, 78–88).** Steepness and quasi-convexity are not equivalent to a positive Hessian determinant or an auxiliary convex function. The rewrite uses a theorem-specific condition.
2. **Blocking — fabricated timescale (24, 111, 854–861).** The numerical exponential shown is about 7.35, not 10^{13} years, and no physical time unit or constants were supplied. This invalidates the stated solar-system benchmark.

3. **Major — exponents (106–109).** "Super-steep" and ϵ-dependent exponents are not a substitute for a cited precise theorem. Exponents cannot be optimized merely to meet a desired age target.
4. **Major — global versus local control.** A truncated Fourier series, short integration, or local Hessian check does not certify the full action domain and time interval. A direct normal-form remainder bound supplies a defensible first milestone and exposes exactly what stronger claims require.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a specified analytic Hamiltonian and action domain, what rigorously computable action excursion and confinement time follow from a certified normal form or quantitative Nekhoroshev theorem?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: one explicit finite-time confinement certificate. Strong: optimized truncation with a stronger bound or full quantitative quasi-convex theorem application. Research extension: exponential scaling across a parameter family; do not infer actual solar-system stability from ϵ alone.

Sources supporting the corrections

- Zhang and Zhang, Improved stability for analytic quasi-convex nearly integrable systems — Theorem-specific quasi-convex exponential stability results.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented

trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.

- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.